MUHAMMAD AKHTAR QUDDUS

Utility CIS, AMI & Smart Metering Consultant · Builder of AI tools for utility data

akhter.quddus@gmail.com | Islamabad, Pakistan | linkedin.com/in/akhterquddus | github.com/akhterquddus-ops

Portfolio: akhterquddus-ops.github.io | Kaggle: kaggle.com/muhammadakhtarquddus

SUMMARY

I have spent more than 17 years on the systems that run electricity utilities: customer information, billing, smart meters, and the field work that connects them. Since 2022 I have led the smart meter mass deployment work at IESCO, where the rollout has passed 1.2 million meters, and worked on CIS and billing for EWA in Bahrain. Before that I spent 13 years at K-Electric, mostly on SAP IS-U. In 2026 I built Utility Data Copilot, an on-premise AI assistant that answers management questions from a utility's own data. I am looking for technical program management roles in energy and utilities.

SKILLS

Utility domain: CIS and billing, AMI and smart metering, meter-to-cash, mass meter deployment, field workforce management

Platforms: Onesait Utilities (Customer, Workforce, Portal, Mobile), SAP IS-U, SAP Fiori, ABAP, Oracle PL/SQL, Microsoft SQL Server

Delivery: Requirements and gap analysis, solution design, UAT, cutover and go-live, end-user training, multi-vendor coordination

AI and data: Python, Streamlit, local LLMs (Ollama), natural-language-to-SQL, RAG, AI evaluation and guardrails, prompt engineering, KPI dashboards

Machine learning: pandas, scikit-learn, feature engineering, cross-validation, error analysis (Kaggle)

EXPERIENCE

Senior Functional Consultant | InfoTech Group (Minsait–Indra partner) Aug 2022 – Present

Islamabad. Client delivery for IESCO (Pakistan) and EWA (Bahrain) on Minsait's Onesait Utilities platform.

- Led the smart meter Mass Deployment System at IESCO (campaign and workforce management): planning campaigns from consumer data, assigning field work, running work orders on mobile and tracking progress. The rollout has passed **1.2 million meters**, about 7 times the original scope.
- Results reported for the programme: IESCO line losses down to **7.6%**, the lowest in Pakistan, and meter tampering close to zero.
- Trained **300+ IESCO staff**, from deputy directors to line superintendents, in 15+ train-the-trainer sessions on 10 modules of the new billing system, including meter reading, collection, debt management and devices.
- Worked on CIS, AMI and billing delivery for EWA in Bahrain at the same time as the IESCO programme.
- Supporting IESCO's move to the new Onesait billing system, now in user acceptance testing.
- Use GitHub Copilot, ChatGPT and Claude to draft specifications and documentation, and check the output against the business rules myself.

Founder | AsaanDigital Applications (own product) Sep 2026 – Present

Utility Data Copilot. Working demo on synthetic data; looking for a first utility pilot.

- Built Utility Data Copilot: a manager types a question in plain English about consumers, billing, collection, losses or smart meters and gets the answer, the SQL behind it and a chart.
- Runs entirely on the utility's own server with a local open-source model and read-only access to SQL Server or Oracle, so no customer data leaves the network.
- Wrote a utility semantic layer (business views, rules such as how collection rate and line loss are calculated, worked examples). On a synthetic test set of 27 questions (23 data questions and 4 that must be refused), it answered **27 of 27** correctly.
- Added a SQL validator that blocked **20 of 20** attack attempts in testing, role-based sign-in, an audit log, and an admin console so non-developers can maintain it.

Founder and Developer | Karobar360 (personal project) 2025 – Present

- Designed and built Karobar360, a free Android app for shopkeepers and households to keep their khata, installments, committees and daily expenses. Released on Google Play in August 2026.
- Features include voice entry in Urdu or English, automatic logging of transaction SMS, and on-device storage so financial data stays on the phone.
- Still early: 5.0 rating from its first 11 reviews. Building it taught me mobile development and LLM integration end to end.

K-Electric Mar 2009 – Jul 2022

Karachi. Pakistan's largest private power utility.

Analytics Manager (Apr 2022 – Jul 2022)

- Built KPI dashboards and management reports for senior leadership, bringing billing, collection and operational data into one consistent view.

IT Specialist / Deputy Manager, SAP IS-U (Mar 2009 – Apr 2022)

- Worked on 20+ SAP IS-U projects end to end, from blueprint to go-live and support.
- Built custom ABAP programs, SAP Fiori apps and OData/REST services for billing, device management and process automation.
- Maintained the integrations between IS-U and the PM, MM and FICA modules.
- Wrote technical documentation, trained end users, and handled production support for billing and device issues.

Early career | Si3, Soorty Enterprises, British American Company (London), RightSolution.net 2004 – 2009

- ERP and software development roles: Oracle forms, reports and database development, requirements gathering, UAT and implementation.

SELECTED PROJECTS

Utility Rules Copilot · CSM Help Desk | github.com/akhterquddus-ops/utility-rules-copilot

- Answers questions from the NEPRA Consumer Service Manual in English or Roman Urdu, citing the exact clause. Right clause found for **33 of 36** test questions, with 0 invented clause references; exact calculators come straight from the CSM tables.

text-to-sql-starter | github.com/akhterquddus-ops/text-to-sql-starter

- Open-source reference implementation: semantic layer, SQL safety validator, read-only execution and evaluation harness. With a local 7B model the semantic layer took accuracy from **4/20 to 16/20**; the validator blocked **21/21** SQL attacks; 57 automated tests.

Titanic ML (Kaggle) | github.com/akhterquddus-ops/titanic-ml

- End-to-end machine-learning workflow (EDA, feature engineering, cross-validation, overfitting diagnosis, error analysis). A family/ticket group-survival feature lifted the Kaggle score to **0.80143**.

CERTIFICATIONS AND RECOGNITION

- SAP Certified Development Associate, SAP Fiori Application Developer (2021)
- Oracle9i PL/SQL Developer Certified Associate (2004)
- “Prompt Perfectionist”, Masters of Craft certificate, InfoTech Group AI Prompt Battle 2026

EDUCATION

Bachelor of Information Technology (Computer Programming), Iqra University, Karachi, 2000 – 2005